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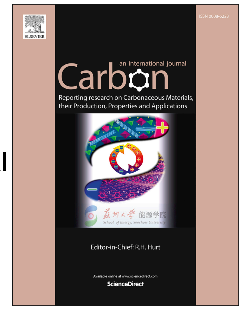

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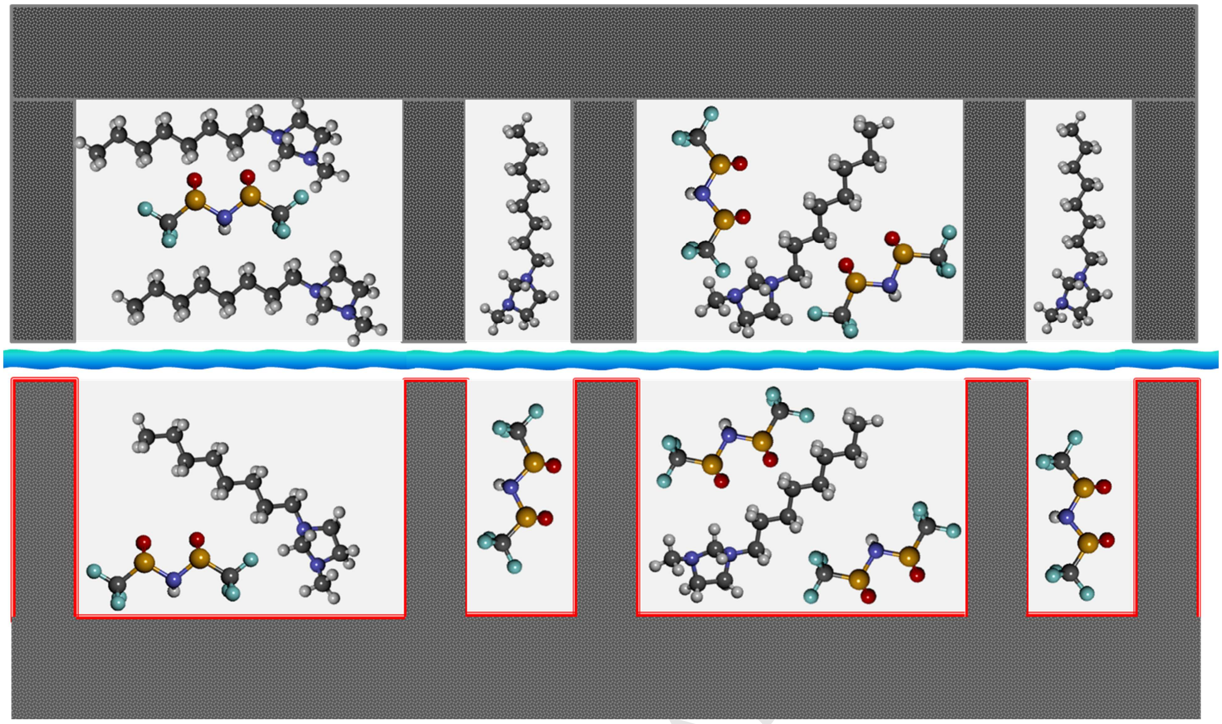

Ionic Liquid Structure, Dynamics, and Electrosorption in Carbon Electrodes with Bimodal Pores and Heterogeneous Surfaces

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Abstract

We investigate the aggregation, diffusion, and resulting electrochemical behavior of ionic liquids inside carbon electrodes with complex pore architectures and surface chemistries. Carbide-derived carbons (CDCs) with bimodal porosities and defunctionalized or oxidized electrode surfaces served as model electrode materials. Our goal was to obtain a fundamental understanding of room-temperature ionic liquid ion orientation, mobility, and electrosorption behavior. Neat 1-octyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide confined in CDCs was studied using an integrated experimental and modeling approach, consisting of quasielastic neutron scattering, small-angle neutron scattering, X-ray pair distribution function analysis, and electrochemical measurements, which were combined with molecular dynamics simulations. Our analysis shows that surface oxygen groups increase the diffusion of confined electrolytes. Consequently, the ions become more than twice as mobile in oxygen-rich pores. Although greater self-diffusion of ions translates into higher electrochemical mobilities in oxidized pores, bulk-like behavior of ions dominates in the larger mesopores and increases the overall capacitance in defunctionalized pores. Experimental results highlight strong confinement and surface effects of carbon electrodes on electrolyte behavior, and molecular dynamics simulations yield insight into diffusion and capacitance differences in specific pore regions. We demonstrate the significance of surface defects on electrosorption dynamics of complex electrolytes in hierarchical pore architectures of supercapacitor electrodes.

Keywords: Energy storage; carbide-derived carbon; ionic liquid; neutron scattering; self-diffusion; supercapacitor; pair distribution function; surface chemistry; molecular dynamics; interface.

Introduction

Supercapacitors combine high energy and power densities, and find use in a broad range of electrical energy storage applications[1]. The fundamental charge/discharge process relies on the electrosorption of ions (from an electrolyte) onto electrode surfaces under applied electric potential [2, 3]. Although classical double layer theory has effectively described this process for most electrode-electrolyte systems, [4] numerous recent discoveries show significant deviations in some systems. In particular, researchers have begun to investigate the effects of confinement in narrow pores [5], ion solvation shell strength [6], screening capabilities of co-ions and counterions [7], diffusion limitations, quantum capacitance contributions [8, 9], etc. Most fundamental modeling and experimental studies have focused on the size ratio of electrode pores to respective electrolyte ions. Since matching ion/pore diameters have yielded optimal charge packing in confinement and highest charge storage densities in supercapacitors (over 200 F g^{-1}) [10], ongoing efforts must tailor electrolytes to appropriate porous electrodes for maximum capacitance, rate handling abilities, and operating voltage windows [11].

To date, most computational analyses of electrosorption have relied on coarse-grained approximations of ions and tailored the simulated porosities of corresponding electrodes to match the electrolyte dimensions. This approach fails to take into account the complex dimensions and chemical properties of ions. In particular, neat (solvent-free) room-temperature ionic liquids (RTILs), which expand the stability windows of supercapacitors [12, 13], include large functional group chains and adopt non-spherical configurations in confined states [14]. The sizes of anions and cations are often mismatched, and carbon electrodes with a single, well-defined pore diameter may not be optimal for co-ion and counterion electrosorption. Furthermore, the chemical behavior of ions is neither metallic nor uniform: imidazolium-based ionic liquids, such as [$EMIm^+$][$TFSI^-$] may feature hydrophobic cations and hydrophilic anions (with strong dipole moments). Therefore, tailoring electrolyte ions to electrode pores may require non-trivial approaches, including complex pore architectures with multiple pore dimensions. Such an approach might facilitate ion partitioning and higher packing densities.

Such an approach will need to investigate the electrosorption of asymmetric ions with different dimensions into hierarchical porous architectures. Few studies have analyzed the state and correlations of ions in confinement, factored in the effects of electrode-electrolyte interactions, or attempted to reconcile experimental characterization with computational modeling. Scattering measurements based on neutron or synchrotron X-rays can provide length- and time-scale information with high levels of atomic resolution and molecular motions. A comprehensive analysis, which is complemented by computational modeling approaches, can provide insight into ion orientation, densification, and dynamics in confinement. Prior studies of RTIL confined in small pores have described substantial densification of ions confined in narrow pores [15, 16]. The electrosorbed layers of co-ions and counterions, orientations of molecules, and both localized (1-2 ion layer length scales) and overall pore filling densities strongly depend on the interactions of specific ions with electrode surfaces. Furthermore, the attractive or repulsive interaction energies between ions and pores control the equilibrium electrosorption densities of ions on surfaces, which, in turn, influence electrochemically driven charge saturation and pore filling/defilling behaviors [17]. To date, simulations and experiments have only partially explored the effects of pore size on overall electrode ionophilicity/ionophobicity [18], and other important variables, such as the structure and surface chemistries of electrodes, which may significantly influence electrostatic interactions and resulting energy storage performance, are also poorly understood.

Carbide-derived carbons (CDCs), which are synthesized *via* Cl_2 etching of metal carbides at 300 – 1,200 ºC, allow precise tuning of subnanometer pore size distributions, making them model systems for fundamental electrosorption experiments [19]. Furthermore, CDCs exhibit robust surface chemistries and disordered carbon structures. Synthesis conditions influence both properties [20]. Most reported supercapacitor studies rely on CDCs synthesized from silicon carbides or titanium carbides, since these precursors offer high specific surface areas (over 1,500 $m^2 g^{-1}$) [21] and monodisperse pore diameters ($d_{average}$ = 0.6 – 0.8 nm) that match desolvated ions of high-performance organic electrolytes [22]. However, ionic liquid molecules are typically non-spherical and larger in size [23], so these very small pore sizes may not be adequate for unrestricted mobility, self-correlation, and diffusion. While high Cl_2 treatment temperatures yield larger pores in most CDCs, they also increase pore size polydispersity and, therefore, impede precise control over ion packing and ion partitioning. Previously, we have demonstrated the important impacts of defects and heterogeneity of electrode-electrolyte interfaces on capacitance [24], ion dynamics [25], and electrochemical stability [26], on simple electrode and electrolyte systems. Most reported CDC model systems used subnanometer pores with uniform (singular) dimensions and narrow diameter distributions [2]. Furthermore, sizes of electrosorbed ions closely matched those of pore diameters, and the simple molecular structures of cations and anions allowed for straightforward dimensional approximations. In contrast, carbonaceous electrodes exhibit broad pore size distributions [16, 20, 27-29], and ionic liquids may incorporate long side chains that induce steric limitations and preclude coarse-grain approximations [30-32]. Steric host-guest interactions play important roles in a variety of electrical charge storage systems, i.e., supercapacitors (electrosorption only), pseudocapacitors (rapid surface reactions), and Faradaic, intercalation chemistry batteries [33]. Therefore, subsequent research efforts must address these real-life parameters and determine the influence of surface heterogeneity on electrochemical behavior of complex, large ions in hierarchical porous carbon networks.

In this work, we integrate multiple experimental techniques with computational simulations to assess behaviors of large ions inside bimodal pore architectures, and evaluate the role of pore surface chemistries. CDCs with bimodal porosities were derived from molybdenum carbide (Mo_2C-CDC), which, due to the large dimensions of the molybdenum metal, yields larger (yet well-controlled) pores [34, 35]. We rely on vacuum annealing and air oxidation to, respectively, remove surface defects [36] or introduce common chemical functionalities onto the pore surfaces. We tailored the bimodal porous model system to an ionic liquid with asymmetrically sized charges: while the [$TFSI^-$] anion could fit into both micropores and mesopores, the larger imidazolium-based cation exhibited more restricted translational and rotational motion. Quasielastic neutron scattering (QENS) and small angle neutron scattering (SANS) characterization approaches were used to measure the densification and mobility (self-diffusion) of the confined RTIL molecules. X-ray total scattering/pair distribution function (TS/PDF) measurements determined ion-ion correlations for both bulk and confined states. We developed a novel molecular dynamics (MD) model of the system and analyzed the behavior and charge accumulation of ions in the differently functionalized slit pores. Electrochemical testing revealed how ion-ion correlations and agglomerates in pores with divergent chemistries translated to capacitance and electrosorption dynamics.

2. Experimental

2.1 Electrode Material Synthesis

Y-Carbon, Inc. (Bristol, PA, USA) supplied the initial CDC, which had been synthesized from molybdenum carbide (Mo_2C) precursor according to an established procedure [34, 35]. Cl_2 gas was used to etch the Mo_2C particles (~ 1.0 µm diameter) at 800 ºC for 5 hours, followed by NH_3 gas annealing at º600 C for 2 hours [37]. The Mo_2C-CDC powder was then loaded into a graphite furnace (Solar Atmospheres, USA) and annealed under a high vacuum ($2\cdot10^{-6}$ tor) *via* the following stepwise thermal treatment: 1) outgas at 25 ºC for 24 hours; 2) ramp-up to 120 ºC at 2 ºC min^{-1} rate; 3) hold at 120 ºC for 1 hour; 4) ramp-up to 400 ºC at 5 ºC min^{-1} rate; 5) hold at 400 ºC for 2 hours; 6) ramp-up to 700 ºC at 8.5 ºC min^{-1} rate; 7) hold at 700 ºC for 2 hours; 8) ramp up to 1,300 ºC at 10 C min^{-1} rate; 9) hold at 1,300 ºC for 14 hours; 10) cool down to room temperature. This material is referred to as “defunctionalized Mo_2C-CDC” [26]. 50% of that powder was loaded into a quartz tube furnace, heated under Ar to 300 °C, and treated with flowing dry air for 5 hours at that temperature. It is referred to as “oxidized Mo_2C-CDC.”

2.2 *Materials Characterization*

A Quadrasorb gas sorption analyzer (Quantachrome Instruments, USA) was used to measure specific surface area (SSA) and porosity through N_2 gas adsorption at isothermal 77 K conditions, with liquid nitrogen as the bath coolant. Brunauer-Emmett-Teller (BET) SSA was calculated from the 0.05 – 0.30 P/P_0 adsorption range [38]. Quenched solid Density Functional Theory (QSDFT) used the adsorption branch [39] to model the pore size distribution and derive surface area and pore volume [40].

A Thermo Fisher Scientific K-Alpha XPS with monochromatic Al K-alpha x-rays was used to analyze surface chemistry and functional group content of defunctionalized and oxidized Mo_2C-CDCs. The instrument used a monochromated, micro-focusing, Al K_α X-ray source (1486.6 eV) with a variable spot size (30-400 µm). Analyses derived average surface compositions with a 400 µm X-ray spot size for maximum resolution and greatest areal coverage. Data was collected and processed using the Thermo Scientific Avantage XPS software package (v 4.61). Spectra were charge corrected using the C1s core level peak set to 284.8 eV.

2.3 Quasi-elastic Neutron Scattering

To prepare samples for neutron scattering measurements, the Mo_2C-CDCs pores were filled with 1-octyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([$OMIm^+$][$TFSI^-$]. Vacuum infiltration introduced the ions to each material to occupy 100% of its gas sorption-measured pore volume. Details of this procedure, as well as the RTIL synthesis method, are described in the Supplementary Content (SI) section. The molecular structure of the RTIL and the predicted dimensions of the ions are shown in Fig. S1 in SI.

The backscattering spectrometer (BASIS) at the Spallation Neutron Source of Oak Ridge National Laboratory (Oak Ridge, TN, USA) was used to study the dynamics of [$OMIm^+$][$TFSI^-$] confined in both defunctionalized and oxidized Mo_2C-CDCs [41]. The Si (111) analyzer crystals, which featured a Bragg reflection at 88°, selected the final neutron wavelength λ_0 = 6.267 Å and provided an averaged energy resolution of 3.5 μeV. Data were taken in the -100 to +100 µeV energy transfer range. The corresponding momentum transfer (Q) spanned from a 0.30 $Å^{-1}$ minimum to 1.9 $Å^{-1}$ maximum. Spectra were collected at 300 K and at 10 K (for resolution) for each sample. The temperatures were controlled using a standard top loading Closed Cycle Refrigerator (CCR). Since hydrogen has the largest incoherent neutron scattering cross-section among the elements that had been present in the materials of this study, and only [$OMIm^+$] includes this element, measured dynamics were dominated by the RTIL cations. The dynamic structure factor, S(Q,E), was obtained after vanadium normalization of the QENS data, and used for diffusion coefficient calculations. Data were reduced using Mantid software [42] and analyzed using DAVE [43] software. The measured curves of scattering intensity, $I(Q,E)$ were approximated by:

$$I(Q,E) = \left[X_1(Q)\delta(E) + \left(1 - X_1(Q)\right)S(Q,E) + B(Q,E)\right] \otimes R(Q,E) \quad , \qquad (1)$$

where $X_1(Q)$ is the elastic scattering of immobile species, and $\delta(E)$ is a Dirac delta function that accounts for the elastic scattering at zero energy transfer. B(Q,E) and R(Q,E), respectively, account for the background and instrument resolution.

2.4. Small Angle Neutron Scattering

SANS data were collected using the EQ-SANS instrument at Oak Ridge National Laboratory (ORNL) [44]. Data were collected at sample-to-detector distances of 1.3 m using a minimum wavelength setting of 2 Å and 4.0 m using a minimum wavelength setting of 10 Å. In both cases the choppers of the instrument ran at 60 Hz. These two settings combined covered an effective Q range of ~ 0.0035 $Å^{-1}$ to 1.4^{-1} $Å^{-1}$, where Q is related to the scattering angle (2θ) and neutron wavelength (λ) by Q = (4π sin(θ))/λ. The SANS measurements were performed at 300 K and 353 K. Data were reduced to correct for sample transmission, background, and scattering from the empty cell using the standard procedures implemented in the Mantid software package.[42] The reduced data were then azimuthally averaged to produce a 1D profile I(Q) vs Q.

2.5 X-ray Total Scattering and Pair Distribution Function Analysis

X-ray TS/PDF measurements were completed at the 11-1D-B beamline of the Advanced Photon Source (Argonne National Laboratory). Pure CDC (defunctionalized and oxidized) powders, bulk $[OMIm^+][TFSI^-]$ liquid, and RTIL confined in each CDC (prepared the same way as for the neutron scattering experiments) were loaded into polyimide capillaries. Scattering measurements were performed in transmission mode at ambient conditions using an incident photon energy of 58.65 keV (λ = 0.2114 Å) and a Perkin Elmer amorphous silicon image plate detector [45] with 15 minute total collection time per sample. Parasitic scattering from the empty polyimide container (duplicating the assembly without a sample) and a Ni powder standard (99.99%, Sigma Aldrich) were measured using the same setup. The program Fit2D was used to calibrate the sample to detector distance and detector alignment using a CeO_2 powder standard [46]. Raw scattering data was integrated into Q-space spectra, and the process applied a mask and polarization correction during integration. The normalized total scattering structural factor, S(Q), was produced in PDFgetX2 by subtracting polyimide container scattering, analyzing the appropriate sample composition [47], and applying standard corrections for the area detector setup [48]. These S(Q) patterns are given in Figure S6 in SI. The corresponding pair distribution function of the sample data, PDF G(r), was calculated *via* a Fourier transformation of the S(Q). The calculations used a Q range interval of 0.5 $Å^{-1}$ (Q_{min}) to 22.2 $Å^{-1}$ (Q_{max}) for all datasets, and are discussed below. The relationship between S(Q) and G(r) is expressed as:

$$G(r) = \frac{2}{\pi}\int_{Q_{min}}^{Q_{max}} Q[S(Q)-1]\sin(Qr)dQ \qquad (2)$$

2.6 Molecular Dynamics

We performed MD simulations in the canonical constant molecule number (N), constant volume (V), and constant temperature (T), or NVT, ensemble using GROMACS [49]. As illustrated in Figure S2 in SI, the simulation box consists of two slit pores immersed in the ionic liquids. The slit pore has a length of 12 nm in the axial direction. Each pore is separated by 7 nm from the neighboring pore, which simulated a reservoir between the electrodes of sufficient size that allowed unconfined electrolyte ions to reach bulk behavior in the middle of the reservoir. Oxidized CDCs were modeled by adding –OH groups to the surface layers of slit pores. During the simulations, the carbon atoms of the electrode were fixed, while functional groups were allowed to rotate and bend. Net charges were uniformly distributed over the electrode surface atoms, generating different potentials between cathode and anode, where negative and positive, respectively, charges were applied to maintain charge neutrality of the system. The electrochemical potentials (V) of the slit surfaces were computed by previously described customized code [50]. A series of surface charges were applied to cover the potential range from 0 V to over 4 V, which is the typical operational voltage window for ILs. A cutoff of 1.1 nm were used for both van der Waals interactions and electrostatic interactions in real space, and long-range electrostatic interactions were calculated by the PME method with a grid spacing of 0.1 nm and a fourth-order inter [51]. Each simulation with charged surfaces was initialized at 800 K for 2 ns and was subsequently annealed to 300 K over 8 ns. After another 10 ns of equilibration at 300 K, a final 20 ns production run was carried out. These simulations were repeated 3 times with different initial configurations to reduce statistical errors. Our prior work includes additional details regarding the force fields and simulation setup [24]. Self-diffusion coefficients of ions were calculated using the Einstein equation. Specifically, for ions confined in slit pores, only the mean-square displacement (MSD) in the directions parallel to the surface were calculated. For these simulations to calculate ion diffusivities confined in slit pores, the production runs were extended to 360 ns. We fit MSD curve to a linear curve to obtain the diffusion coefficients according to the Einstein formula:

$$D = \lim_{t\to\infty} \frac{1}{4t} \langle [r(t) - r(0)]^2 \rangle \quad (3)$$

2.7 Electrochemical Characterization

Using a previously described procedure,[52, 53] carbon powders were compressed into thin electrode films. A 60 wt. % dispersion of PTFE in H_2O (Sigma-Aldrich) was mixed with excess ethanol and Mo_2C-CDCs to create 95 wt. % carbon / 5 wt. % binder slurries, which were, subsequently, ground in agate mortar and pestle for uniform consistency. A mechanical rolling mill was used to produce 120 µm thick freestanding electrode films (2.8 mg cm^{-2} mass loading).

Electrodes were punched into 12 mm diameter disks and assembled into sandwiches (pouch cells), with carbon-coated aluminum current collectors (Exopak) and 2 layers of 3501 Celgard® separator between each electrode. The cells used three-electrodes in classical configuration, i.e., a small working electrode (WE) and an oversized counter electrode (CE) composed of commercially available activated carbon. Neat $[OMIm^+][TFSI^-]$ was used as the electrolyte for all experiments, and all electrochemical measurements were carried out at room temperature. A quasi-reference configuration (symmetrical electrodes, with the reference electrode measuring the potential difference and not setting it, was also used [54]. A chlorinated Ag/AgCl wire (directly submerged in the same neat $[OMIm^+][TFSI^-]$ electrolyte) was used as a reference electrode in each case. All tests were carried out inside of an Ar-filled glovebox with a VMP3 potentiostat (Bio-Logic, Inc., France).

Electrochemical impedance spectroscopy (EIS) was measured in the 200 kHz – 10 mHz dampening oscillatory regime. EIS measurements used a 10 mV amplitude centered at 0.0 V. Cyclic voltammetry (CV) measurements were carried out at 0.5 – 1000 mV s^{-1} sweep rates in the 0.0 ↔ 2.5 V window. SSA-normalized capacitance (*Csp)* was calculated from the discharge segments using the following equation:

$$C_{sp} = \frac{2}{SSA}\int_{V_o}^{V_f} \frac{I}{\frac{dV}{dt} m_{WE} V_W} dV \tag{4}$$

In the above equation, V_0 and V_f denote, respectively, the vertex potentials; m_{WE} is the mass of a single working electrode; *SSA* is the DFT-derived specific surface area of the defunctionalized or oxidized Mo_2C-CDC; V_w is the electrochemical potential window; and $\frac{dV}{dt}$ is the sweep rate.[55] Square wave chronoamperometry (CA), which was conducted using asymmetric cells, applied step voltage from 0.0 V to ±1.25 V, ±1.00 V , ±0.75 V, ±0.50 V, and ±0.25 V. Each CA measurement recorded charge accumulation for 60 minutes, until steady-state conditions were attained.

3. Results

3.1 Pore Structure and Surface Chemistry

Mo_2C-CDCs are comprised of porous carbon networks with high SSAs and complex architectures. As shown in Fig. 1a, N_2 sorption measurements exhibited Type IV isotherm behavior, and demonstrated well-defined hysteresis in the desorption branch. This behavior was expected for CDCs synthesized from Mo_2C precursors, and agreed with previously reported results [34, 35]. The hysteresis shape suggests bottleneck-shaped pores [56]. The pore size distribution plots (Fig. 1b) shows presence of pores with two predominant sizes: micropores (with a pore diameter = 0.79 nm) and mesopores (with a 2.60 nm predominant pore diameter). The larger pores more significantly contributed to the total porosities: they constituted 78% of the total pore volume of defunctionalized Mo_2C-CDCs and 80% of the total pore volume of oxidized Mo_2C-CDCs. As summarized in Table 1, both materials exhibited very high SSAs (2,040 – 2,374 m^2 g^{-1}) and pore volumes (1.77 – 2.16 cm^3 g^{-1}).

The SANS data for the defunctionalized, RTIL-free ("empty") Mo_2C-CDCs were analyzed with the polydisperse hard sphere (PDSP) model implemented in the PRINSAS software [57]. Although the SANS data indicates significant microporosity and/or surface roughness, contributions from such excessively large features are outside of the resolution range of the SANS instrument (Fig. 1c). The analysis revealed a significant fraction of mesopores with a well-defined 2.6 nm mean pore diameter and a narrow pore size distribution (Fig. 1d). The pore size histogram shows no presence of larger mesopores. The PDSP model, which also correlates internal surface area with pore dimensions, shows that micropores accounted for up to ~90% of the total internal surface area, while the mesopores contributed 10% of internal SSA.

While the overall porosity and bimodal nature of the pore distribution closely matched the N_2 adsorptive behavior and DFT modeling results, the results from these techniques differed in their assessment of relative prevalence of micropores and mesopores. As previous analyses have shown, small-angle scattering techniques detected and accounts for all pores, including closed pores and internal pockets, whereas gas sorption could only interrogate accessible, open-ended pores [58]. Furthermore, while the PDSP model for SANS assumes spherical pore shapes, the QSDFT models rely on slit pore shape assumptions to achieve best model fits [39]. These key differences, combined with the inherent measurement uncertainties of both techniques, along with principal difficulties in distinguishing surface roughness from micropores, limit our interpretation and explain the obtained micropore to mesopore ratio differences between the two techniques.

The air oxidation treatment was specifically tuned to retain this well-ordered bimodality and, indeed, did not substantially restructure the material. Oxidation increased the SSA of Mo_2C-CDCs by 16%, which translated to negligible activation and an unchanged pore size distribution. However, this treatment changed the surface chemistry of the materials by doubling the number of oxygen-containing species. XPS elemental analysis of the surface (summarized in Table S1 in SI) revealed that, after 300 ºC air treatment, oxygen content increased from 1.1 at.% to 2.2 at.%. Functional group deconvolution (shown in Fig. S3 in SI) reveals that most added surface species were in forms of C=O and O–C=O groups. Since the materials were, initially, annealed under high vacuum at 1,300 ºC, they were highly graphitized.[36] Subsequently, XPS analysis showed substantial π-π bonding (indicative of sp^2 hybridization and high structural order in the carbon material) for both defunctionalized and oxidized Mo_2C-CDC. Surface functionalization did not alter the graphitization of the porous carbons, a finding that is also supported by X-ray TS/PDF data (discussed below).

3.2. Ion Dynamics and Pore Filling Densities

QENS results were obtained from porous CDCs filled with $[OMIm^+][TFSI^-]$ using the vacuum infiltration approach**.** The procedure yielded near-complete pore filling of both micropores and mesopores, with only 0.002 $cm^3\ g^{-1}$ void volume left in both defunctionalized and oxidized Mo_2C-CDCs (shown in Fig. S4 in the SI). Note that QENS results reflect the dynamics of hydrogen-containing species of $[OMIm^+]$ cations. The QENS spectra exhibited a substantially non-Lorentzian behavior, which warranted use of the Cole-Cole distribution for the modeling of the scattering function S(Q,E) ,[59] :

$$S(Q,E) = \frac{1}{\pi E_0(Q)}\left[\frac{\left(E/E_{0(Q)}\right)^{-\alpha(Q)}\cos\frac{\pi\alpha(Q)}{2}}{1+2\left(E/E_{0(Q)}\right)^{1-\alpha(Q)}\sin\frac{\pi\alpha(Q)}{2}+\left(E/E_{0(Q)}\right)^{2(-\alpha(Q))}}\right] \tag{5}$$

The width parameter $E_0(Q)$ is analogous to HWHM (half width at half maximum) of a Lorentzian peak, and α(Q) is a parameter describing deviation from a simple Lorentzian spectral shape, with $0 \leq \alpha \leq 1$. If $\alpha = 0$, equation 3 represents a Lorentzian function. A value of α other than zero is due to a distribution of relaxation times, leading to non-Lorentzian character of QENS data. Modeling results using Eq. 3 for $[OMIm^+][TFSI^-]$ confined in defunctionalized and oxidized Mo_2C-CDCs are shown in Fig. S5 (SI). The obtained values for $E_0(Q)$ (Eq. 3) were then used to calculate self-diffusion coefficients (D_t) and jump lengths, L, for jump diffusion processes (Eq. 6), (Eq. 7):

$$E_0(Q) = \frac{D_t Q^2}{1 + D_t Q^2 \tau_0}, \tag{6}$$

$$D_t = \frac{L^2}{6\tau_0} \tag{7}$$

The width parameter $E_0(Q)$ increased with Q at low Q values, and flattened at higher Q values, which indicates translational jump diffusion behavior. The obtained diffusion coefficients and jump diffusion lengths are summarized in Table 2.

The QENS spectra and Q-resolved width parameters for both defunctionalized and oxidized Mo_2C-CDC are shown in Fig. 2. The diffusion coefficient, obtained from the low-Q slopes of the plots in Fig. 2b, was higher for the RTIL confined in oxidized Mo_2C-CDC than for RTIL confined in defunctionalized CDCs. More precisely, self-diffusive mobility of [$OMIm^+$][$TFSI^-$] in oxidized pores was 2.6 times larger as compared to defunctionalized pores. This finding agrees with our previous results, which had demonstrated improvement of similar imidazolium-based RTILs mobility in functionalized pores [25]. Since the bimodal CDCs possess sufficiently large pores that exceed the dimensions of the both cations and anions, they allow translational and rotational, as well as longer-range random walk, motions. Although ions are expected to move in pairs, it is unclear whether the large cations (with a long hydrophobic tail) or the smaller anions are the main drivers of mobility.

The increased mobility of RTIL ions in oxidized Mo_2C-CDC surface, which contains more hydrophilic –O–C=O– and C=O groups, may be attributed to more complete "wetting" of the pore surfaces by the ionic liquid molecules. Oxidized surfaces exhibit attractive intermolecular interactions with the polar groups of the [$TFSI^-$] anion and draw more ions closer to the pore surfaces. Consequently, oxidized pores contain fewer ions in centers of pores (i.e. not directly bound to pore surfaces) than defunctionalized pores. This arrangement leaves more room for ions to move around each other and makes them more mobile. A similar concept had described the fast diffusion of ionic liquids confined in mesoporous carbon [60].

The fraction of elastic scattering (shown in Fig. S5 in SI), which assessed the fraction of zero energy transfer events during neutron-molecule collisions, consistently showed 6 – 14% higher elasticity for ions confined in oxidized pores than for $[OMIm^+][TFSI^-]$ in defunctionalized CDCs. Since each system included approximately the same number of hydrogens, this could have only stemmed from different intermolecular interaction strengths between the RTIL and pore surfaces. In the case of oxidized surfaces, more molecules strongly attached to pore walls. Subsequently, they followed the CDC backbone's dynamics, which was devoid of diffusion-like mobility, leading to more elastic neutron scattering from cations in the sample. When surfaces were devoid of functional groups, more loosely bound ions more readily engaged in diffusion-like dynamics, yielding a broader QENS signal. Therefore, oxidized pore surfaces were associated with a greater number of surface-bound cations, but higher mobilities of those cations that had not been tightly bound to the surface.

Small-angle neutron scattering (SANS) data were collected from the IL-filled defunctionalized and oxidized pores at 300 K as well as from an unfilled, defunctionalized CDC sample to show the impact of the IL on the signal as presented in Fig. 3. Fig. 3a shows data as obtained, whereas Fig. 3b compares the scattering from net confined ionic liquid with that of the "empty" defunctionalized CDC carbon structure. Although the surface chemistry had little effect on the SANS data from the $[OMIm^+][TFSI^-]$ filled samples at $Q < 0.03Å^{-1}$, where scattering profiles overlapped with each other, both samples strongly contrasted against the unfilled CDC. All of the measured samples had a kink at low Q ($Q \cong 0.008$ $Å^{-1}$); it was more prominent in samples filled with $[OMIm^+][TFSI^-]$. This effect is an instrument artifact and is highly dependent on the sample composition, particularly the hydrogen content [61]. The intermediate Q (0.03 $Å^{-1}$ to 0.1 $Å^{-1}$) regime revealed some differences that could be attributed to different amounts of $[OMIm^+][TFSI^-]$ penetration into the CDCs. The scattering intensities were higher from $[OMIm^+][TFSI^-]$ confined in the oxidized CDCs when compared to those from the defunctionalized pores. Once the ionic liquid started to fill the pores, the scattering contrast (scattering length density (SLD) difference between the CDCs and the ionic liquid), which is directly related to the scattering intensity, decreased. The higher amount of the ionic liquid inside the CDCs pores resulted in a lower scattering contrast, and, therefore, demonstrated lower scattering intensities. The SLD of Mo_2C-CDC was calculated to ~ 6.66 x 10^{-10} $Å^{-2}$ (assuming ~2 g cm^{-3} density) and that of the ionic liquid (density = 1.32 g cm^{-3}) is 1.46 x 10^{-10} $Å^{-2}$.[13] The presence of a greater amount of hydrogen-bearing species in the oxidized pores was supported by the higher intensities observed for Q > 0.2 $Å^{-1}$, which was dominated by the incoherent background scattering from the hydrogen in the samples.

The SANS curves of $[OMIm^+][TFSI^-]$-filled CDC (Fig. 3b) showed a characteristic shoulder near Q = 0.07 $Å^{-1}$, which indicates the presence of scattering structures of net confined RTIL at that length scale. Assuming spherical shape of these scattering objects, their size may be estimated by the following approximation: $R \approx 3/Q$. It yielded a dimension of ~9 nm for the scattering features. The presence of these aggregates indicated that, while the experimental procedure (and subsequent gas sorption and QENS measurements) expected 100% filling of pore volumes, the $[OMIm^+]TFSI^-]$ did not uniformly occupy the entire available void space in the pores. Subsequently, a more accurate representation of RTIL-filled pores accounts both empty spaces (with no ions) and agglomerates (which exhibit greater densities than the ionic liquid's bulk ~1.32 g cm^{-3} value). These features were absent in empty Mo_2C-CDC pores, because their incoherent background scattering levels (by H atoms) were very different, and the low-Q data changed considerably. This analysis reinforces our earlier studies, which had demonstrated significant pore densification of RTIL in carbon micropores.[15] Based on the porosimetry analysis of the RTIL-filled CDCs, these aggregates possibly formed in the pore spaces near the grain surfaces and effectively blocked N_2 access into the partially unfilled internal pore domains. We speculate that, in an effort to minimize interfacial energy and these clusters formed patches of RTIL tightly attached to pore surfaces and maximized their agglomerated densities in the process.

The data fitting, performed over a Q-range of 0.01Å^{-1} to 0.5Å^{-1}, employed a sum of a power law model and the Debye-Anderson-Brumberger (DAB) model [62], which was previously employed for modeling SANS data from nanoporous CDC,[63] and is shown in Eqn. 8.

$$I(Q) = \frac{A}{Q^{\alpha}} + \frac{B\xi^3}{[1+(Q\xi)^2]^2} + C \tag{8}$$

The factor A is the scale factor for the power law model, while α is the power law exponent. This model is well-suited to fitting data from fractal, highly polydisperse and aggregate structures. The second term is the DAB model. Here, ξ described the characteristic length scale between the two interpenetrating phases, while and B was the DAB scaling constant. The constant C accounted for the incoherent background scattering of the sample. The results of the fitting performed using this model are presented in Table 3.

The value of α was roughly the same (~4), which suggests smooth surfaces, for both [$OMIm^+$][$TFSI^-$] filled samples. This contrasts with scattering from empty CDCs that derived an α ~2.4 and revealed a rough aggregate structure. The values of ξ were lower in the oxidized, [$OMIm^+$][$TFSI^-$] filled CDC as compared to defunctionalized RTIL-filled CDC sample. These parameter values were consistent with expected lower penetration of the RTIL into the pores relative to the oxidized CDC. The most significant change in the fitting results for the oxidized and defunctionalized CDCs was found in the DAB scaling factor (B) expressed as a fraction of the sum of the three scaling/constant terms (A, B, and C) that exist in equation 6 as this measure will be more independent of the amount of material in the beam, which can be difficult to control for powders. This ratio was very high in unfilled CDC and decreased by almost 2 orders magnitude for $[OMIm^+][TFSI^-]$-filled CDCs. Since more IL molecules occupied defunctionalized pores than oxidized pores (according to the SANS results), $[OMIm^+][TFSI^-]$-filled defunctionalized CDCs reduced the scattering length the most and, subsequently, produced the lowest DAB scaling ratio.

In aggregate, SANS results suggest that the more hydrophilic surfaces of the oxidized CDCs attracted more anions of the $[OMIm^+][TFSI^-]$ to the pore surfaces and reduced the number of the RTILs molecules in the center of the pores, which, subsequently, created a more mobile environment with greater molecular self-diffusion. This interpretation agreed with the quasi-elastic scattering results. Together, the present results confirm a previously proposed explanation for similar observations in several related systems of ionic liquids confined in CDC pores with different surface chemistries.[24, 25]

3.3 X-Ray Pair Distribution Function Analysis of Empty and RTIL-Filled CDCs

We relied on X-Ray pair distribution function (PDF) data to contrast changes in ion-ion correlations between bulk and confined states of RTIL in CDCs. The most informative approach is done through the examination of the net difference PDFs $\Delta G(r)$, which subtracted empty CDC scattering from each IL-filled CDC configuration: $\Delta G(r) = G(r)_{IL\text{-}filled\text{-}CDC} - G(r)_{empty\text{-}CDC}$ (Fig. 4).

Direct X-ray PDF analysis (plotted as r*G(r) vs. r) showed nearly-identical structures for both empty CDCs and no influence of surface chemistry on the degree of graphitization (Fig. 4a and 4b). The total PDF was dominated by the structural correlations of nanoporous carbon, which matched previously described diverse local C–C ordering.[45, 64, 65] Since surface functional groups (e.g., C=O bonds) constituted a relatively low fraction of total material (based on XPS characterization), low elemental sensitivities of X-ray scattering technique did not allow any discernable signals from these moieties. In comparison, the total PDFs of each RTIL-filled CDC exhibited extra intensities that augmented signals from their corresponding empty carbon counterparts (Fig. 4c and 4d). The first peak at ~1.4 Å in RTIL-filled CDC data became much more intense due the presence of C–C and C–N bonds (in [$OMIm^+$]), and C–F, C–S and S–N, and S=O bonds (in [$TFSI^-$]) from IL molecules (all have similar bonding distances).

The $\Delta G(r)$ analyses highlight ion-ion correlation differences between states of bulk (unconfined liquid) and confined (pore-filled) [$OMIm^+$][$TFSI^-$]. While pore surface chemistry did not alter the structure of the confined RTIL (within analysis errors), the confinement process significantly altered the properties of the ionic liquid. The broad and intense ‘bump,’ which had been featured at ~15-24 Å in all confined RTIL PDF samples, was absent in bulk RTIL PDF.

3.4 *Molecular Dynamics Simulation*

In order to further identify the origin of the variations seems in the difference PDFs, a classical molecular dynamics (MD) model was developed using a simplified 2.6 nm slit pore morphology (approach described in section 2.5). We benchmarked bulk IL experimental PDF with simulated PDF from MD model (Fig. 5a) to compare experimental and simulated PDFs of ions confined in pores with defunctionalized and oxidized surfaces. Structural properties of force-field generated bulk IL demonstrated good qualitative agreement with the experimental observations and correctly described the shape/oscillations of the long-r range features (Fig. 5a bulk RIL curve). For confined ions, the simulated PDFs also accurately ehiibited the 'bump' features at ~15-24 Å (Fig. 5a), albeit narrower in width than their experimental counterparts. This was likely due to the simpler single-size slit geometry used in the MD calculation. Further investigations of surface chemistry effects were made possible *via* partial PDF analysis (Fig. 5b). This approach split the total PDF into a sum of three partials: [$TFSI^-$]-[$TFSI^-$], [$OMIm^+$]-[$OMIm^+$], and [$OMIm^+$]-[$TFSI^-$] correlations. The [$OMIm^+$]-[$TFSI^-$] correlation clearly oscillated in an opposite phase in contrast to the other two correlations. Comparisons between each set of partial PDFs reveals distinguishable, surface chemistry-specific interactions between ions and surfaces. For instance, oxidized surfaces shifted [$OMIm^+$]-[$TFSI^-$] correlations toward the high-r region.

An interesting takeaway from the MD-generated PDFs was the fact that, in agreement with experimental results, surface chemistry did not alter cumulative ion behavior. However, this description of ion-ion correlations is independent from their spatial orientations inside of pores. Since PDF's diffraction method measures ensemble average of all intermolecular ion-ion correlations, it can neither separate partial correlations from the experimental data nor distinguish between ion-ion and ion-surface interactions. Given the complexity of this system, PDF helped us benchmark against molecular dynamics simulations. This was vital for our efforts to extract predictive information about the ion orientation, ion-surface interactions, and dynamics of ions in confinement.

Fig. 6a shows the number density distribution inside of 2.6 nm diameter slit pores at the point of zero charge (PZC). Base on peak locations, we divided the plots into 2 regions. Region 1 accounted for the condition of strong ion adsorption, which significantly increased the number density over the bulk density. In this region, as compared to defunctionalized pores, oxidized pores showed more affinity to [$TFSI^-$] ions than to [$OMIm^+$] ions ([$TFSI^-$] peaks were higher and closer to the surface, while the [$OMIm^+$] peaks were lower and more dispersed). While ions assumed a similar layered structure in region 2, the number density was only slightly higher than the bulk density. Subsequently, oxidized pores demonstrated less prominent ion peaks than defunctionalized pores in this region. The accumulated ion number density is shown in Table 4. The MD results closely corresponds with the SANS results described above: even though the oxidized pores attracted more ions, especially the [$TFSI^-$] to the surface, it contains fewer ions in the middle of the pore, and the overall number of ions inside is lower than in the pristine pore.

As shown in Fig. 6b and 6c (angles formed by the vector connecting 2 C atoms and the normal vector of the slit surface), these regions influenced ion orientations. In region 1, due to π–π interactions, the imidazolium ring in cations more favorably arranged parallel to the electrode surfaces, which accounted for the location of first [$OMIm^+$] peak nearest the electrode surface (Fig 6b). The significant angle distribution of [$TFSI^-$] around 90° also suggested anions oriented themselves along electrode surfaces in region 1 (Fig. 6c). Oxidized surfaces intensified both effects. While region 2 featured more random ion configurations, oxidized surfaces induced more perpendicular [$OMIm^+$] orientations in pore centers.

The diffusion coefficients of ions in different regions were calculated and are shown in Fig. 7a. Confinement of ions in slit pores noticeably decreased the overall diffusion coefficients, and smaller distances between ions and surfaces further continued that trend. Electrode-ion interactions strongly attached surface ions to pore interfaces and prevented facile diffusion (Fig. S7 in SI). Furthermore, high ion densities near the wall impeded ion mobility out of their respective shells. Oxidized surfaces made this effect more significant: diffusion coefficients in oxidized pores were nearly two orders of magnitude lower in region 1 than in defunctionalized pores and one order of magnitude lower in region 2. These results correlated with the QENS-derived observations that ions were more strongly attached to the oxidized surface, which yielded more elastic scattering.

In an effort to more thoroughly explain the QENS results, we adopted a similar strategy to define the observable mobility of ions in slit pore models. First, this approach calculated the mean-square displacement (MSD) of each ion. For a given Q, an ion was considered to be mobile if its MSD, within a certain time period, exceeded $(\pi/Q)^2$. Second, an average MSD of all mobile ions yielded the diffusion coefficients *via* a linear regression fit within the range of Q-values that had been accessible to QENS. Higher Q values captured more ion species (Fig. 7b). As illustrated in Fig. 7c, at lower Q, oxidized pores yielded lower diffusion coefficients than those in defunctionalized pores. This trend reversed at high Q, where the ion diffusion coefficients in oxidized pores were actually higher. This result further confirms our previous explanation that, while oxidized surfaces strongly attract ions, this ionophilicity creates space for fast diffusion of ions in pore middles.

We modeled specific capacitance (Fig. 7d) using the following equation:

$$C_{SP} = \frac{\sigma_{anode} - \sigma_{cathode}}{\phi_{anode} - \phi_{cathode}} = \frac{2\sigma}{\Delta\phi} \tag{9}$$

The calculation used σ as the surface charge density and $\Delta\phi$ as potential difference between anode and cathode. The simulation predicted lower capacitance in oxidized 2.6 nm diameter pores, which may be related to reduced ability of functionalized pores to both accommodate ions and efficiently reorganize them into dense configurations.[66] To contrast the behavior of ions in mesopores with that in micropores, we also attempted to simulate [$OMIm^+$][$TFSI^-$] in 0.8 nm slit pores. However, the resulting diffusion coefficients were found to be negligible (Fig. S7c in SI). Computational studies yielded a one order of magnitude slower diffusion of ions in nanoconfined environments than in the bulk.[67] These results agree with prior studies that showed similar poor diffusion of confined ions in electrodes with neutral or slight surface charge.[68] Furthermore, during the charging/discharging process, ions in nanoporous electrodes exhibited sloshing dynamics in slit channels [69] and an uneven distribution during rapid charging dynamics.[70] Therefore, the slow mobilities of ions in 0.8 nm slit pores preclude significant all-atom MD simulation analyses.

3.5 Electrochemical Behavior of [OMIm$^+$][TFSI$^-$] in Bimodal CDCs

Electrochemical impedance measurements are shown in Fig. 8. The Nyquist plot comparison (Fig. 8a) showed higher [OMIm$^+$][TFSI$^-$] mobilities under a dynamic potential field in the presence of surface oxygen groups. Despite exhibiting near-identical equivalent series resistance (R_s) parameters, the mid-range ionic impedance was 8.1 Ω higher for defunctionalized CDCs than for oxidized CDCs. Although impedance was almost identically capacitive-dominated at 10 mHz (-81.9º for oxidized CDCs and -80.9º for defunctionalized CDCs), oxidized CDCs were more predominantly capacitive at mid-to-high frequencies (Fig. 8b). The equilibrium time constant Fig. 8c) was 27% lower for the RTIL in oxidized Mo_2C-CDC pores, which suggested more rapid electrosorption processes for electrodes with oxygen-rich surfaces.

These findings, which were driven by electrochemical processes, paralleled the derived self-diffusion behavior of QENS measurements (under ambient electrochemical conditions) and showed the significant influence of intrinsic electrode interfaces on ion electrosorption dynamics. Oxygen surface groups made the CDC surfaces more ionophilic and ensured more efficient [OMIm$^+$][TFSI$^-$] mobilities alongside pore walls.

Although gravimetric capacitance (shown in Fig. S8a in SI) for oxidized CDCs (76 F g^{-1} at 10 mV s^{-1}) exceeded the C_{sp} for defunctionalized CDCs (73 F g^{-1} at 10 mV s^{-1}), the SSA-normalized capacitance for defunctionalized CDCs was greater for this electrode-electrolyte system (Fig. 9a). These results agreed with the MD simulation. Larger pores made more bulk space available and facilitated free ion mobility and exchange during electrosorption with fewer contacts with the surfaces. Subsequently, while [$TFSI^-$] ions likely arranged themselves in parallel orientation with respect to the hydrophilic (ionophilic) pore walls, hydrophobic [$OMIm^+$] ions positioned themselves as far away from surfaces as possible and minimized total charge densities in oxidized pores. However, some narrow channels (the subnanometer pore components) impeded ion mobilities in the defunctionalized CDCs.

Cyclic voltammograms at low sweep rates (Fig. 9b and 9c**)** showed inflection points during charge/discharge processes around +0.25 V. Figure inserts highlight this effect. Greater capacitance at lower potentials, followed by a dip at that voltage, and, subsequently, by an increase in capacitance at greater potentials signified possible charge saturation and surface chemistry-dependent partial ion de-filling.[17] The charge accumulation and integral capacitance of oxidized CDCs was greater for this low-voltage portion of the CV (calculations shown in Fig. S8b in SI). Furthermore, a drop in capacitance at low potentials was more pronounced for the defunctionalized CDC and underscored its ionophobic pore-electrolyte interface. The effect became more “smeared” at higher charge/discharge rates (Fig. 9d). The full summary of electrochemical performance and key parameters is shown in Table 5. A three-electrode measurement with a quasi-reference Ag/AgCl electrode,[54] which is shown in Fig. S9 in SI, suggested saturation and 0.8 nm pore de-filling behavior for both [$OMIm^+$] (in the negative potential regime) and [$TFSI^-$] (in the positive potential regime). Previous experiments have confirmed codependent, unidirectional mobilities of both cations and anions during electrosorption.[71] Therefore, any pore de-filling process involved both co-ions and counterions.

Square wave static accumulation measurements (Fig. 10a) further demonstrated differences in potential-dependent charge saturation of oxidized and defunctionalized CDCs. During instantaneous voltage jumps from 0.00 V to +1.25 V and -1.25V (Fig. 10b**)**, oxidized pores required less time to achieve charge saturation and demonstrated time vs. charge inflection points at ~200 seconds, whereas defunctionalized pores steadily accumulated charge throughout the entire square wave process. After 3,600 seconds of accumulation time at 1.25 V, defunctionalized Mo_2C-CDCs 6.2 stored μC cm^{-2}, as opposed to 5.4 μC cm^{-2} for oxidized CDCs. However, during smaller voltage jumps from 0.00 V to +0.25 V and -1.25V, charge accumulation in defunctionalized pores demonstrated an inflection point at ~70 seconds (Fig. 10c), and oxidized pores ended up storing a nearly identical amount of charge (0.02 μC cm^{-2} more) as the defunctionalized Mo_2C-CDCs. This voltage-dependent difference in behavior paralleled the performance of cyclic voltammograms at similar potentials.

Surface oxidation influenced the electrochemically-driven mobilities of $[OMIm^+][TFSI^-]$ ions inside of bimodal CDCs. While the micropores were sufficiently large and relatively easily accommodated anions, the large dimensions of the cations precluded total rotational and translational mobilities of these molecules in the narrow pores. Prior to electrochemical cycling, the small (0.8 nm diameter) pores were expected to contain low concentrations of both cations and anions; "electrowetting" of these pores likely occurred only during high voltages.[72] While ion electrosorption dynamics (which were higher in oxidized pores) more heavily influenced capacitance at low voltages, influence of total ion accumulation densities (which were higher in defunctionalized pores) became more dominant at higher potentials. The results resembled previously reported pore saturation and de-filling measurements,[17] as well as continuum modeling simulations of this electrosorption behavior.[5, 18]

The results underscore the intricate effects of pore size and surface chemistry on capacitance and ion dynamics. On the one hand, in agreement with QENS-derived ion diffusion measurements, surface oxidation improved the mobilities of the IL. This finding agrees with previous results, which had shown higher ionic mobilities in functionalized CDC pores.[25] On the other hand, this advantage did not translate into greater overall C_{sp} of oxidized pores. This takeaway, in turn, agrees with previous results that demonstrated higher capacitance on defunctionalized surfaces in non-porous electrodes (with no ion confinement).[24] While oxygen groups improved the [TFSI$^-$]'s interface with pore surfaces, they repelled the alkyl chain on the [OMIm$^+$] cations. In all likelihood, the net effect of this process reduced the number density of electrosorbed ions on accessible surfaces. Since the narrow pores did not provide sufficient degree of freedom for both translational and rotational diffusion of cations, ions inside the mesopores contributed most to the QENS signal.[73, 74] Subsequently, even in larger pores with more bulk-like ion dynamic behaviors, oxidized pore surfaces improved pore ionophilicities and influenced electrolyte dynamics. Since the mesopores accounted for 78-79% of the total porosity of each sample, they constituted a larger fraction of the total capacitance.

4. Conclusions

Surface functional groups significantly influenced the self-organization, mobility, and electrochemically-driven dynamics of [OMIm$^+$][TFSI$^-$] room-temperature ionic liquid inside of bimodal Mo_2C-CDC pores. Functionalization of pore surfaces, which had been initially graphitized and defect-free, increased self-diffusion of the confined molecules by 150%.

Molecular dynamic simulations further confirmed enhanced ion mobilities in the cores of oxidized mesopores. While ion-to-ion correlations remained constant in differently functionalized pores, RTIL ions demonstrated substantial differences in coordination and arrangement between their bulk and confined states. Electrochemical measurements agreed with neutron scattering results and demonstrated the positive influence of oxidized pore interfaces on ion dynamics. Oxidized surfaces reduced the electrochemical ionic impedance of $[OMIm^+][TFSI^-]$ by over 50%. However, due to the bimodal pore structure of Mo_2C-CDCs, the partial contributions of ions from the totally confined (microporous) and partially confined/semi-bulk (mesoporous) regions yielded a net negative effect of oxidized surfaces on capacitance. Although oxygen groups attracted more ions closer to pore walls, vacated the centers of narrow channels, and improved IL mobilities, they reduced electrosorption charge accumulation densities by over 10%. MD simulations of charge accumulation identified similar behaviors.

Our efforts examined the effects of heterogeneity and disorder at electrode-electrolyte interfaces of complex systems that resemble emerging supercapacitors. Porous carbons with hierarchical porosities offer very high surface areas, and large ionic liquid molecules demonstrate exceptional electrochemical stability under high applied potentials. We showed the extent to which oxygen surface moieties affect interactions between ions and pores, notwithstanding the significant effects of micro- and mesopore-induced confinement of large, irregular molecules. We correlated the properties of confined electrolytes under neutral potentials with their electrosorption behavior and capacitance under dynamic potential conditions. We successfully applied pair distribution functional analysis to decouple ion-ion interactions from their enveloping carbon matrix, highlight differences between bulk and confined states, and develop computational models that correctly described both the correlation behavior and resulting charge storage densities. These techniques will be essential for future studies of energy storage materials and interfaces.

Intermolecular interactions between pores surfaces with different surface chemistries and electrolyte molecules significantly affected ion mobilities and capacitance. Although pore size has, to date, been predominantly used to evaluate and optimize ion confinement effects, we demonstrated the applicability of surface heterogeneity and disorder at the electrode-electrolyte interface for pore saturation and ion partitioning studies. Furthermore, we showed that specific parameters of electrochemical experiments, such as magnitudes of applied potentials and impedance studies, reveal different aspects of fundamental behavior of large electrolytes in confined pores. The behavior of these ions in very narrow confinement remains difficult to characterize using neutron scattering technique, model using computational approaches, or isolate from overall electrochemical data. The complex hierarchical pore structures resemble a transmission line model, and the heterogeneity and intermolecular interactions at the electrode-electrolyte interface add further intricacy to this system. Future experimental and computational techniques must investigate these systems in greater detail.

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References:

[1] Simon P, Gogotsi Y. Capacitive Energy Storage in Nanostructured Carbon–Electrolyte Systems. Accounts of Chemical Research. 2013;46(5):1094-103.
[2] Béguin F, Presser V, Balducci A, Frackowiak E. Carbons and Electrolytes for Advanced Supercapacitors. Advanced Materials. 2014;26(14):2219–51.
[3] Gogotsi Y. Not just graphene: The wonderful world of carbon and related nanomaterials. MRS Bulletin. 2015;40(12):1110-21.
[4] Conway BE. Electrochemical Supercapacitors: Scientific Fundamentals and Technological Applications. New York: Springer; 1999.
[5] Kondrat S, Wu P, Qiao R, Kornyshev AA. Accelerating charging dynamics in subnanometre pores. Nat Mater. 2014;13(4):387-93.
[6] Pean C, Daffos B, Rotenberg B, Levitz P, Haefele M, Taberna P-L, et al. Confinement, Desolvation, And Electrosorption Effects on the Diffusion of Ions in Nanoporous Carbon Electrodes. Journal of the American Chemical Society. 2015;137(39):12627-32.
[7] Kondrat S, Kornyshev A. Superionic state in double-layer capacitors with nanoporous electrodes. Journal of Physics: Condensed Matter. 2011;23(2):022201.
[8] Zhan C, Neal J, Wu J, Jiang D-e. Quantum Effects on the Capacitance of Graphene-Based Electrodes. The Journal of Physical Chemistry C. 2015;119(39):22297–303.
[9] Wood BC, Ogitsu T, Otani M, Biener J. First-Principles-Inspired Design Strategies for Graphene-Based Supercapacitor Electrodes. The Journal of Physical Chemistry C. 2014;118(1):4-15.
[10] Chmiola J, Largeot C, Taberna P-L, Simon P, Gogotsi Y. Desolvation of Ions in Subnanometer Pores and Its Effect on Capacitance and Double-Layer Theory. Angewandte Chemie. 2008;120(18):3440-3.
[11] Frackowiak E, Beguin F. Carbon materials for the electrochemical storage of energy in capacitors. Carbon. 2001;39(6):937-50.
[12] Fedorov MV, Kornyshev AA. Ionic Liquids at Electrified Interfaces. Chemical Reviews. 2014;114(5):2978-3036.
[13] Hayes R, Warr GG, Atkin R. Structure and Nanostructure in Ionic Liquids. Chemical Reviews. 2015;115(13):6357-426.
[14] Canongia Lopes JNA, Pádua AAH. Nanostructural Organization in Ionic Liquids. The Journal of Physical Chemistry B. 2006;110(7):3330-5.
[15] Bañuelos JL, Feng G, Fulvio PF, Li S, Rother G, Dai S, et al. Densification of Ionic Liquid Molecules within a Hierarchical Nanoporous Carbon Structure Revealed by Small-Angle Scattering and Molecular Dynamics Simulation. Chemistry of Materials. 2014;26(2):1144-53.
[16] Bañuelos JL, Feng G, Fulvio PF, Li S, Rother G, Arend N, et al. The influence of a hierarchical porous carbon network on the coherent dynamics of a nanoconfined room

temperature ionic liquid: A neutron spin echo and atomistic simulation investigation. Carbon. 2014;78:415-27.
[17] Mysyk R, Raymundo-Piñero E, Béguin F. Saturation of subnanometer pores in an electric double-layer capacitor. Electrochemistry Communications. 2009;11(3):554-6.
[18] Kondrat S, Kornyshev AA. Pressing a spring: what does it take to maximize the energy storage in nanoporous supercapacitors? Nanoscale Horizons. 2016.
[19] Gogotsi Y, Presser V. Carbon nanomaterials. 2nd ed: CRC Press; 2013.
[20] Schütter C, Ramirez-Castro C, Oljaca M, Passerini S, Winter M, Balducci A. Activated Carbon, Carbon Blacks and Graphene Based Nanoplatelets as Active Materials for Electrochemical Double Layer Capacitors: A Comparative Study. Journal of the Electrochemical Society. 2015;162(1):A44-A51.
[21] Dash R, Chmiola J, Yushin G, Gogotsi Y, Laudisio G, Singer J, et al. Titanium carbide derived nanoporous carbon for energy-related applications. Carbon. 2006;44(12):2489-97.
[22] Chmiola J, Yushin G, Gogotsi Y, Portet C, Simon P, Taberna PL. Anomalous Increase in Carbon Capacitance at Pore Sizes Less Than 1 Nanometer. Science. 2006;313(5794):1760-3.
[23] Aken KLV, McDonough JK, Li S, Feng G, Chathoth SM, Mamontov E, et al. Effect of cation on diffusion coefficient of ionic liquids at onion-like carbon electrodes. Journal of Physics: Condensed Matter. 2014;26(28):284104.
[24] Dyatkin B, Zhang Y, Mamontov E, Kolesnikov AI, Cheng Y, Meyer HM, et al. Influence of Surface Oxidation on Ion Dynamics and Capacitance in Porous and Non-Porous Carbon Electrodes. The Journal of Physical Chemistry C. 2016.
[25] Dyatkin B, Mamontov E, Cook KC, Gogotsi Y. Capacitance, Charge Dynamics, and Electrolyte-Surface Interactions in Functionalized Carbide-Derived Carbon Electrodes. Progress in Natural Science - Materials International. 2015;25:631–41.
[26] Dyatkin B, Gogotsi Y. Effects of Structural Disorder and Surface Chemistry on Electric Conductivity and Capacitance of Porous Carbon Electrodes. Faraday Discussions. 2014;172:139-62.
[27] Tsai W-Y, Gao P-C, Daffos B, Taberna P-L, Perez CR, Gogotsi Y, et al. Ordered mesoporous silicon carbide-derived carbon for high-power supercapacitors. Electrochemistry Communications. 2013;34:109-12.
[28] Wang D-W, Li F, Liu M, Lu GQ, Cheng H-M. 3D Aperiodic Hierarchical Porous Graphitic Carbon Material for High-Rate Electrochemical Capacitive Energy Storage. Angewandte Chemie International Edition. 2008;47(2):373-6.
[29] Noked M, Soffer A, Aurbach D. The electrochemistry of activated carbonaceous materials: past, present, and future. Journal of Solid State Electrochemistry. 2011;15(7-8):1563-78.
[30] Perkin S, Salanne M, Madden P, Lynden-Bell R. Is a Stern and diffuse layer model appropriate to ionic liquids at surfaces? Proceedings of the National Academy of Sciences. 2013;110(44):E4121-E.
[31] Canongia Lopes JN, Deschamps J, Pádua AAH. Modeling Ionic Liquids Using a Systematic All-Atom Force Field. The Journal of Physical Chemistry B. 2004;108(6):2038-47.
[32] Olivier-Bourbigou H, Magna L. Ionic liquids: perspectives for organic and catalytic reactions. Journal of Molecular Catalysis A: Chemical. 2002;182–183:419–37.
[33] Dyatkin B, Halim J, Read J. Electrode Surface Composition of Dual-Intercalation, All-Graphite Batteries. C. 2017;3(1):5.
[34] Janes A, Thomberg T, Kurig H, Lust E. Nanoscale fine-tuning of porosity of carbide-derived carbon prepared from molybdenum carbide. Carbon. 2009;47(1):23-9.

[35] Kim HS, Singer JP, Gogotsi Y, Fischer JE. Molybdenum carbide-derived carbon for hydrogen storage. Microporous and Mesoporous Materials. 2009;120(3):267-71.
[36] Chmiola J, Osswald S, Gogotsi Y. Structural Evolution of Carbide-Derived Carbons upon Vacuum Annealing. Carbon. 2012;50:4880-6.
[37] Portet C, Kazachkin D, Osswald S, Gogotsi Y, Borguet E. Impact of synthesis conditions on surface chemistry and structure of carbide-derived carbons. Thermochimica Acta. 2010;497(1-2):137–42.
[38] Brunauer S, Emmett PH, Teller E. Adsorption of Gases in Multimolecular Layers. Journal of the American Chemical Society. 1938;60(2):309-19.
[39] Caguiat JN, Kirk DW, Jia CQ. Uncertainties in characterization of nanoporous carbons using density functional theory-based gas physisorption. Carbon. 2014;72:47-56.
[40] Gor GY, Thommes M, Cychosz KA, Neimark AV. Quenched solid density functional theory method for characterization of mesoporous carbons by nitrogen adsorption. Carbon. 2012;50(4):1583 –90.
[41] Mamontov E, Herwig KW. A time-of-flight backscattering spectrometer at the Spallation Neutron Source, BASIS. Review of Scientific Instruments. 2011;82(8).
[42] Arnold O, Bilheux JC, Borreguero JM, Buts A, Campbell SI, Chapon L, et al. Mantid-Data analysis and visualization package for neutron scattering and mu SR experiments. Nuclear Instruments & Methods in Physics Research Section a-Accelerators Spectrometers Detectors and Associated Equipment. 2014;764:156-66.
[43] Azuah RT, Kneller LR, Qiu YM, Tregenna-Piggott PLW, Brown CM, Copley JRD, et al. DAVE: A Comprehensive Software Suite for the Reduction, Visualization, and Analysis of Low Energy Neutron Spectroscopic Data. Journal of Research of the National Institute of Standards and Technology. 2009;114(6):341-58.
[44] Zhao J, Gao CY, Liu D. The extended Q-range small-angle neutron scattering diffractometer at the SNS. Journal of Applied Crystallography. 2010;43(5):1068-77.
[45] Smith MA, Foley HC, Lobo RF. A simple model describes the PDF of a non-graphitizing carbon. Carbon. 2004;42(10):2041-8.
[46] Hammersley AP, Svensson SO, Hanfland M, Fitch AN, Hausermann D. Two-dimensional detector software: From real detector to idealised image or two-theta scan. High Pressure Research. 1996;14(4-6):235-48.
[47] Qiu X, Thompson JW, Billinge SJL. PDFgetX2: a GUI-driven program to obtain the pair distribution function from X-ray powder diffraction data. Journal of Applied Crystallography. 2004;37(4):678.
[48] Chupas PJ, Qiu X, Hanson JC, Lee PL, Grey CP, Billinge SJL. Rapid-acquisition pair distribution function (RA-PDF) analysis. Journal of Applied Crystallography. 2003;36(6):1342-7.
[49] Abraham MJ, Murtola T, Schulz R, Páll S, Smith JC, Hess B, et al. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. SoftwareX. 2015;1:19-25.
[50] Feng G, Cummings PT. Supercapacitor Capacitance Exhibits Oscillatory Behavior as a Function of Nanopore Size. The Journal of Physical Chemistry Letters. 2011;22(2):2859–64.
[51] Essmann U, Perera L, Berkowitz ML, Darden T, Lee H, Pedersen LG. A smooth particle mesh Ewald method. Journal of Physical Chemistry. 1995;103:8577–93.

[52] Dyatkin B, Presser V, Heon M, Lukatskaya MR, Beidaghi M, Gogotsi Y. Development of a Green Supercapacitor Composed Entirely of Environmentally Friendly Materials. ChemSusChem. 2013;6(12):2269–80.
[53] Jost K, Stenger D, Perez CR, McDonough JK, Lian K, Gogotsi Y, et al. Knitted and screen printed carbon-fiber supercapacitors for applications in wearable electronics. Energy & Environmental Science. 2013;6(9):2698-705.
[54] Van Aken KL, Beidaghi M, Gogotsi Y. Formulation of Ionic-Liquid Electrolyte to Expand the Voltage Window of Supercapacitors. Angewandte Chemie International Edition. 2015;54(16):4806-9.
[55] Stoller MD, Ruoff RS. Best practice methods for determining an electrode material's performance for ultracapacitors. Energy & Environmental Science. 2010;3:1294-301.
[56] Lowell S, Shields JE. Powder surface area and porosity: Springer Science & Business Media; 2013.
[57] Hinde A. PRINSAS - a Windows-based computer program for the processing and interpretation of small-angle scattering data tailored to the analysis of sedimentary rocks. Journal of Applied Crystallography. 2004;37(6):1020-4.
[58] Laudisio G, Dash RK, Singer JP, Yushin G, Gogotsi Y, Fischer JE. Carbide-Derived Carbons: A Comparative Study of Porosity Based on Small-Angle Scattering and Adsorption Isotherms. Langmuir. 2006;22(21):8945-50.
[59] Cole KS, Cole RH. Dispersion and Absorption in Dielectrics I. Alternating Current Characteristics. The Journal of Chemical Physics. 1941;9(4):341-51.
[60] Chathoth SM, Mamontov E, Dai S, Wang X, Fulvio PF, Wesolowski DJ. Fast diffusion in a room temperature ionic liquid confined in mesoporous carbon. Epl. 2012;97(6).
[61] Do C, Heller WT, Stanley C, Gallmeier FX, Doucet M, Smith GS. Understanding inelastically scattered neutrons from water on a time-of-flight small-angle neutron scattering (SANS) instrument. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment. 2014;737(Supplement C):42-6.
[62] Debye P, Anderson HR, Brumberger H. Scattering by an Inhomogeneous Solid. II. The Correlation Function and Its Application. Journal of Applied Physics. 1957;28(6):679-83.
[63] He L, Chathoth SM, Melnichenko YB, Presser V, McDonough J, Gogotsi Y. Small-angle neutron scattering characterization of the structure of nanoporous carbons for energy-related applications. Microporous and Mesoporous Materials. 2012;149(1):46–54.
[64] Petkov V, DiFrancesco R, Billinge S, Acharya M, Foley H. Local structure of nanoporous carbons. Philosophical Magazine B. 1999;79(10):1519-30.
[65] Forse AC, Merlet C, Allan PK, Humphreys EK, Griffin JM, Aslan M, et al. New Insights into the Structure of Nanoporous Carbons from NMR, Raman, and Pair Distribution Function Analysis. Chemistry of Materials. 2015;27(19):6848-57.
[66] DeYoung AD, Park S-W, Dhumal NR, Shim Y, Jung Y, Kim HJ. Graphene Oxide Supercapacitors: A Computer Simulation Study. The Journal of Physical Chemistry C. 2014;118(32):18472-80.
[67] Zhan C, Lian C, Zhang Y, Thompson MW, Xie Y, Wu J, et al. Computational Insights into Materials and Interfaces for Capacitive Energy Storage. Advanced Science. 2017:1700059-n/a.
[68] He Y, Qiao R, Vatamanu J, Borodin O, Bedrov D, Huang J, et al. Importance of Ion Packing on the Dynamics of Ionic Liquids during Micropore Charging. The Journal of Physical Chemistry Letters. 2016;7(1):36-42.

[69] He Y, Huang J, Sumpter BG, Kornyshev AA, Qiao R. Dynamic Charge Storage in Ionic Liquids-Filled Nanopores: Insight from a Computational Cyclic Voltammetry Study. The Journal of Physical Chemistry Letters. 2015;6(1):22-30.
[70] Pak AJ, Hwang GS. Charging Rate Dependence of Ion Migration and Stagnation in Ionic-Liquid-Filled Carbon Nanopores. The Journal of Physical Chemistry C. 2016;120(43):24560-7.
[71] Richey FW, Dyatkin B, Gogotsi Y, Elabd YA. Ion Dynamics in Porous Carbon Electrodes in Supercapacitors Using in situ Infrared Spectroelectrochemistry. Journal of the American Chemical Society. 2013;135(34):12818–26.
[72] Boukhalfa S, Gordon D, He L, Melnichenko YB, Nitta N, Magasinski A, et al. In-Situ Small Angle Neutron Scattering Revealing Ion Sorption in Microporous Carbon Electrical Double Layer Capacitors. ACS Nano. 2014;8(3):2495–503.
[73] Chathoth SM, Mamontov E, Fulvio PF, Wang X, Baker GA, Dai S, et al. An unusual slowdown of fast diffusion in a room temperature ionic liquid confined in mesoporous carbon. EPL (Europhysics Letters). 2013;102(1):16004.
[74] Chathoth SM, Mamontov E, Dai S, Wang X, Fulvio PF, Wesolowski DJ. Fast diffusion in a room temperature ionic liquid confined in mesoporous carbon. EPL (Europhysics Letters). 2012;97(6):66004.

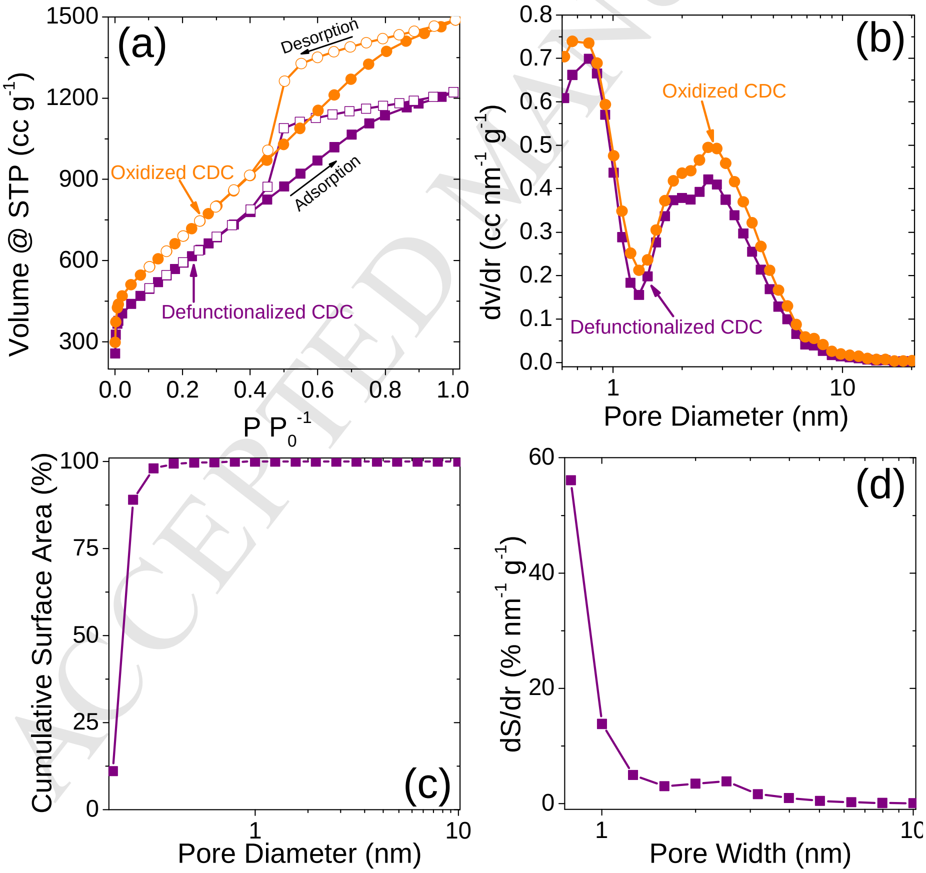

Fig. 1 – (a) N_2 sorption isotherms for defunctionalized and oxidized Mo_2C-CDCs. (b) N_2-sorption derived pore size distribution. (c) SANS-derived pore surface area analysis and a (d) corresponding pore size distribution of the empty, RTIL-free Mo_2C-CDC.

Table 1 – BET and DFT specific surface area and micropore and mesopore volumes of defunctionalized and oxidized bimodal Mo_2C-CDC

Material	BET SSA ($m^2\ g^{-1}$)[a]	QSDFT SSA ($m^2\ g^{-1}$)[b]	Micropore Diameter Mode (nm)[c]	Micropore Volume ($cm^3\ g^{-1}$)[c]	Mesopore Diameter Mode (nm)[d]	Mesopore Volume ($cm^3\ g^{-1}$)[d]	Total Pore Volume ($cm^3\ g^{-1}$)
Defunctionalized CDC	2039.6	1914.8 m^2/g	0.79	0.39	2.60	1.38	1.77 cm^3/g
Oxidized CDC	2373.9	2241.5 m^2/g	0.72	0.44	2.60	1.72	2.16 cm^3/g

[a] Calculated in the 0.05 – 0.30 P/P_0 range
[b] Calculated using the adsorption branch only
[c] Range assumed to account for pores in the 0.61 – 1.30 nm range (first pore size distribution mode)
[d] Range assumed to account for pores in the 1.42 – 34.48 nm range (second pore size distribution mode)

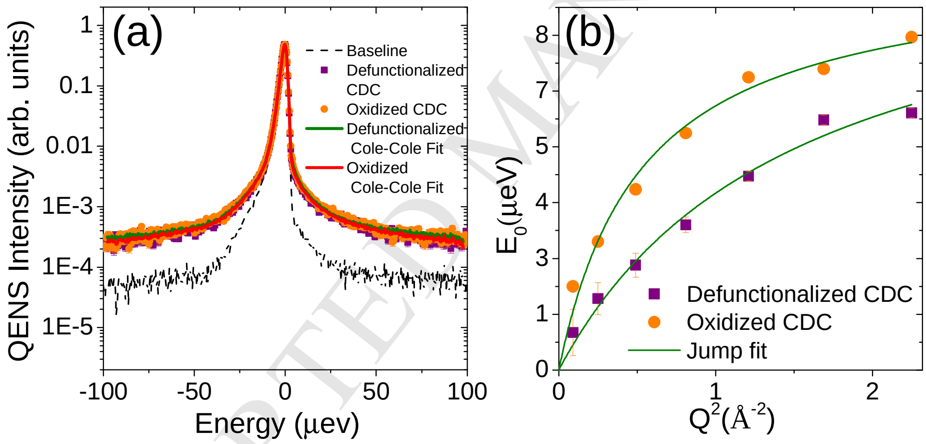

Fig. 2 – (a) QENS spectra for [OMIm$^+$][TFSI$^-$] confined in defunctionalized and oxidized Mo_2C-CDCs at Q = 0.3 Å^{-1}. (b) Dependence of the width parameter of QENS spectra extracted from the Cole-Cole model function plotted as a function of Q-squared. Solid line shows the jump diffusion model fits. The low-Q slope represents the diffusion coefficient.

Table 2 – QENS-derived diffusion coefficients and jump lengths of [OMIm$^+$][TFSI$^-$] confined in defunctionalized and oxidized Mo2C-CDC pores at 300 K

Pore Surface	Diffusion Coefficient ($10^{-10}\ m^2\ s^{-1}$)	Jump length (Å)
Oxidized CDC	2.71 ± 0.32	3.3
Defunctionalized CDC	1.05 ± 0.14	1.9

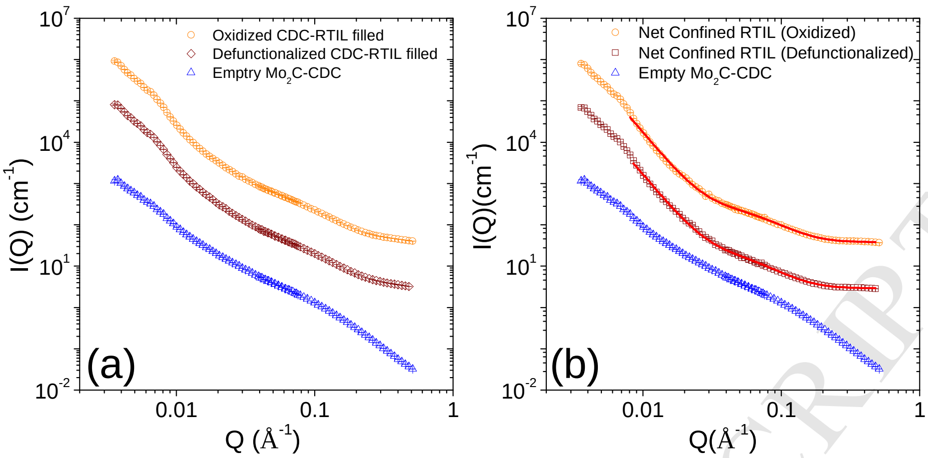

Fig. 3 – Comparison of SANS profiles of $[OMIm^+][TFSI^-]$ confined in oxidized and defunctionalized Mo_2C-CDC with bare annealed CDC measured at 300 K. (a) Spectra of $[OMIm^+][TFSI^-]$ together with both CDCs (b) Spectra from net confined $[OMIm^+][TFSI^-]$ Symbols correspond to the data as specified in the figure and the solid lines are fit as described in the text. Spectra have been shifted vertically for clarity.

Table 3 – SANS-derived fitting parameters for the sum of power law and Debye-Anderson-Brumberger (DAB) model for defunctionalized and oxidized Mo_2C-CDC with and without confined $[OMIm^+][TFSI^-]$

Pore Surfaces	Temp.	$[OMIm^+]$ $[TFSI^-]$	Coefficient (A)[a]	α[a]	DAB scale factor (B)[a]	ξ (Å)[a]	Incoherent scattering (C)[a]	B/(A+B+C)[a]
Defunctionalized (IL-filled)	300 K	Yes	2.1E-6	3.94	0.0014	11.53	0.29	0.0048
Oxidized (IL-filled)	300 K	Yes	2.0E-6	3.95	0.0021	10.64	0.38	0.0054
Defunctionalized (empty)	300 K	No	0.0016	2.38	0.0055	7.27	0.017	0.22

[a] Uncertainties of these fitting parameters are less than 1%

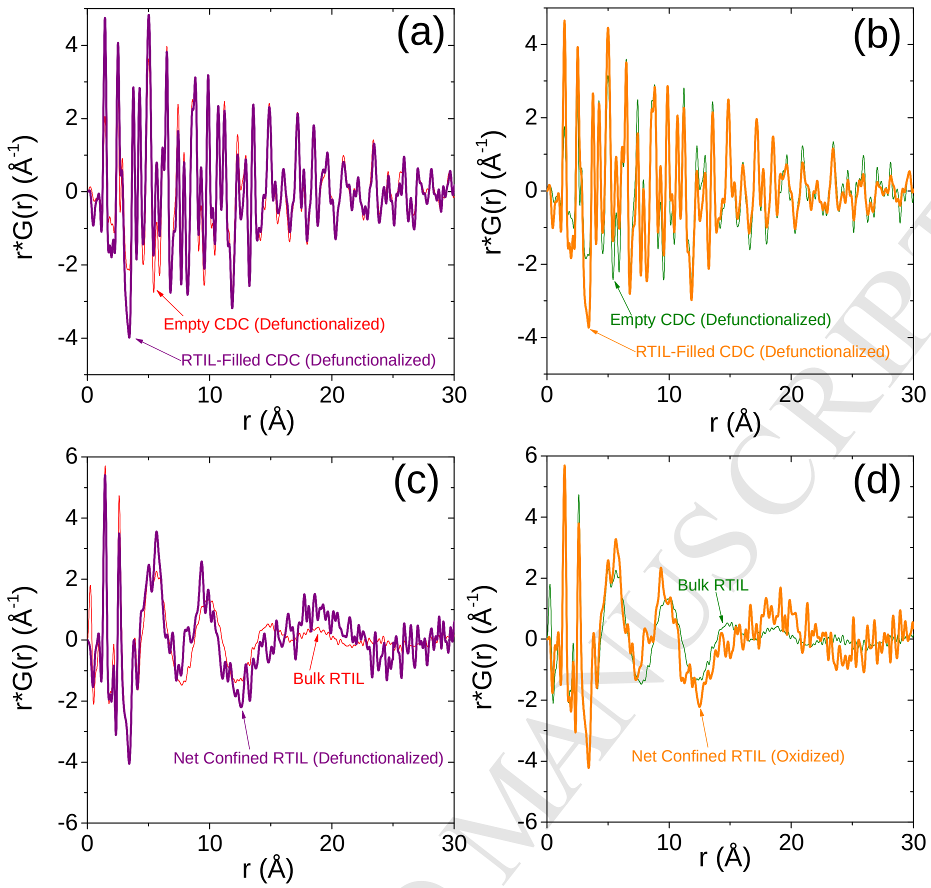

Fig. 4 – (a) Experimental X-ray PDFs of empty CDCs and (b) IL-filled CDC in defunctionalized and oxidized systems. Normalized difference PDFs of confined ILs in comparison with bulk IL data is shown for (c) defunctionalized CDC and (d) oxidized CDC. In all plots, the y-axis is presented as r*G(r) to emphasize signals in the high-r region.

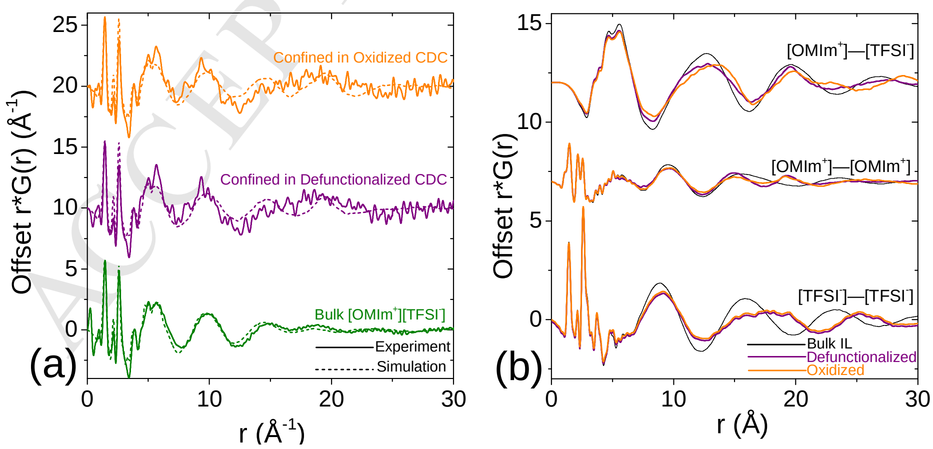

Fig. 5 – (a) MD-generated PDFs of bulk and confined ILs (in defunctionalized and oxidized slit pore) compared against to experimental observation. (b) MD-generated partial PDFs of bulk and confined ILs, showing changes in ion-ion correlations upon the confinement.

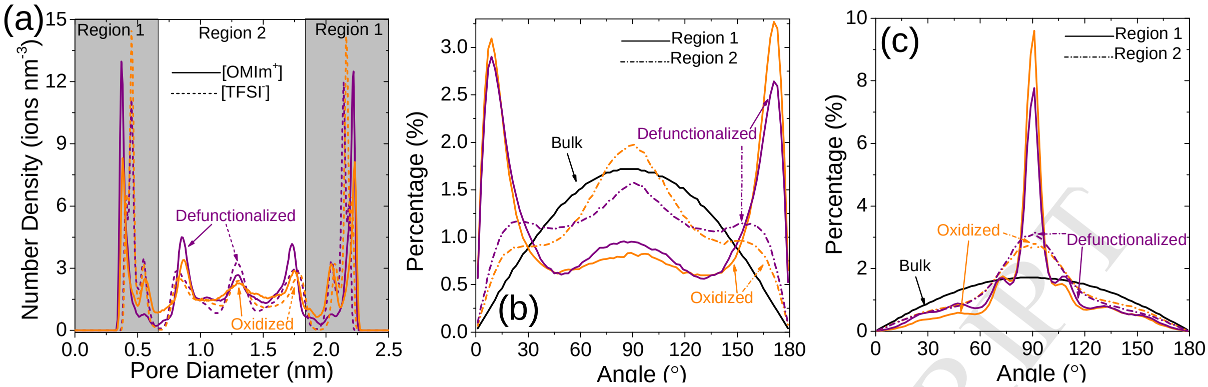

Fig. 6 – (a) Ion number density distribution in the direction perpendicular to the surface inside 2.6 slit pores with neutral surface charge. (b) Angle distribution of [OMIm$^+$] (angle formed by the normal vector of the ring and the normal vector of slit surface) and (c) [TFSI$^-$].

Table 4 – MD-derived accumulated number density (# nm^{-2}) in the direction perpendicular to the surface

	Pristine [OMIm$^+$]	Pristine [TFSI$^-$]	Oxidized [OMIm$^+$]	Oxidized [TFSI$^-$]
Region 1	1.63	1.87	1.64	1.97
Region 2	2.32	2.07	2.22	1.89
Total	3.95	3.94	3.86	3.86

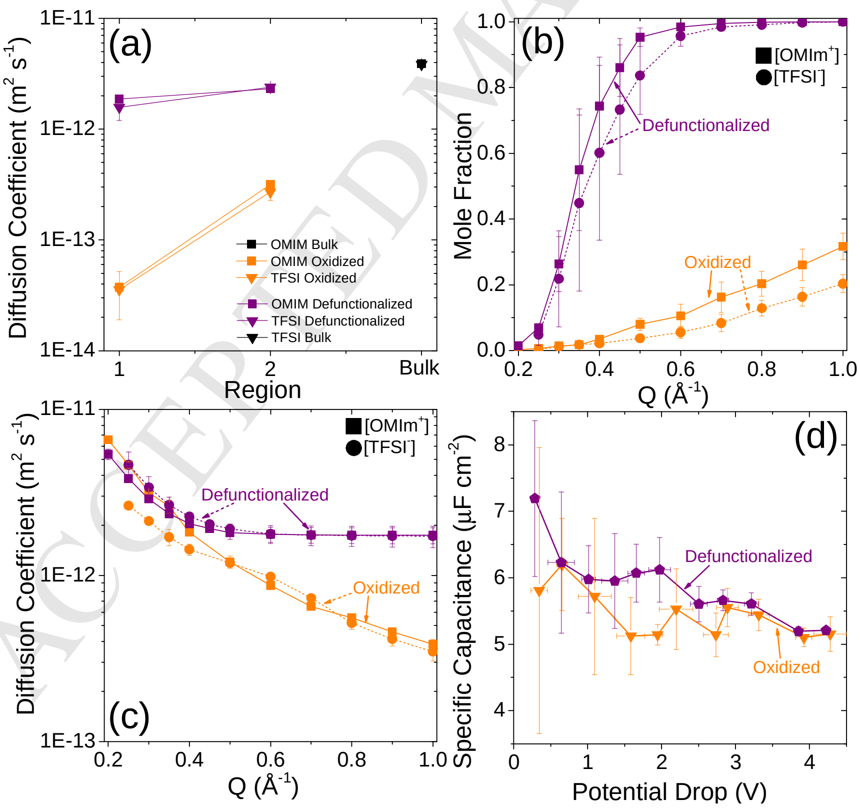

Fig. 7 – (a) Ion diffusion coefficients in different regions and bulk. (b) Mole fraction of selected ions as a function of Q. (c) Diffusion coefficient of selected ions as a function of Q. (d) Specific capacitance as a function of potential.

Table 5 – Summary of Electrochemical Behavior of $[OMIm^+][TFSI^-]$ in defunctionalized and oxidized Mo2C-CDC pores

Pore Surface	R_s (Ω)[a]	Ionic Impedance (Ω)[b]	Time Constant (s)[c]	C_{sp}: 0 ↔ 2.5 V, 0.5 mV s^{-1} (µF cm^{-2})[d]	C_{sp}: 0 ↔ 2.5 V, 10 mV s^{-1} (µF cm^{-2})[d]	C_{sp}: 0 ↔ 0.5 V, 0.5 mV s^{-1} (µF cm^{-2})[d]	C_{sp}: 0 ↔ 0.5 V, 10 mV s^{-1} (µF cm^{-2})[d]
Oxidized	10.2	24.2	6.01 s	4.14	3.39	2.81	1.18
Defunctionalized	11.4	10.5	8.28 s	4.44	3.91	2.37	1.11

[a] Measured from the Nyquist impedance plot, collected from the Re(Z) value at which the Im(Z) value is equal to 0.
[b] Measured from the Nyquist impedance plot, collected from the Re(Z) segment between charge transfer resistance and -45 phase angle
[c] Measured from the Nyquist impedance plot, collected from the frequency that matched a -45 phase angle
[d] Collected from the discharge segments of cyclic voltammetry sweeps

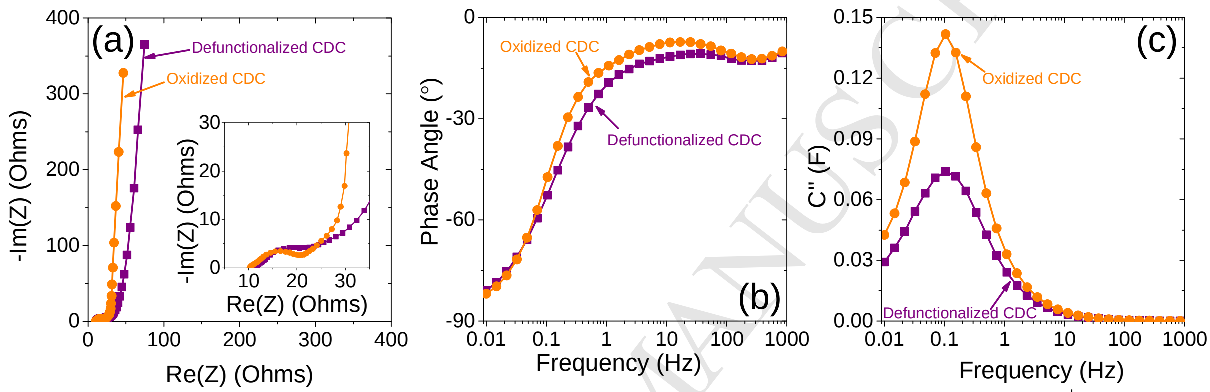

Fig. 8 – (a) Nyquist plot that compares real and imaginary impedance of $[OMIm^+][TFSI^-]$ in defunctionalized and oxidized Mo_2C-CDC pores. (b) Phase angle vs. frequency relationship for electrochemical impedance in the 10^{-2} – 10^3 Hz frequency range. (c) Bode impedance plot that compares the imaginary capacitance component of impedance (C'') vs. oscillation frequency for ionic liquid in defunctionalized and oxidized CDCs.

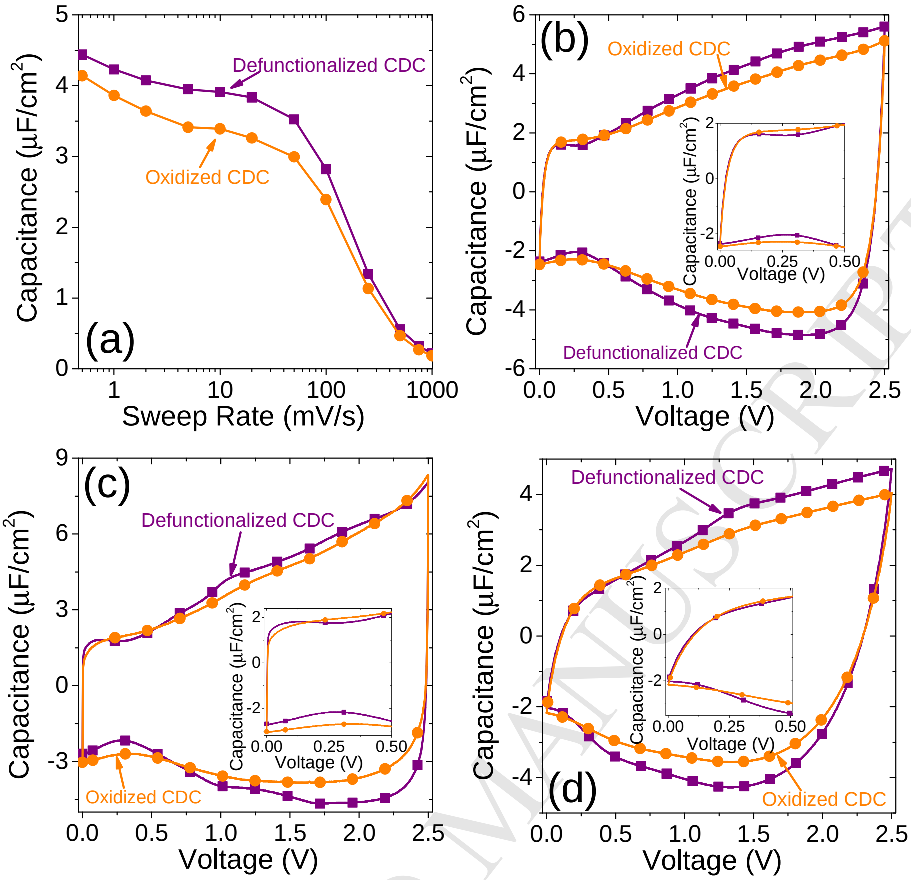

Fig. 9 – (a) Cyclic voltammetry-derived rate handling comparison of SSA-normalized capacitance of $[OMIm^+][TFSI^-]$ in defunctionalized and oxidized Mo_2C-CDC in the 0.5 – 1000 mV s^{-1} sweep range and a 0 ↔ 2.5 V voltage window. Specific cyclic voltammograms are shown for (b) 0.5 mV s^{-1}, (c) 10 mV s^{-1}, and (d) 50 mV s^{-1} sweep rates. The insets in each figure provide zoomed-in 0 ↔ 0.50 V window range.

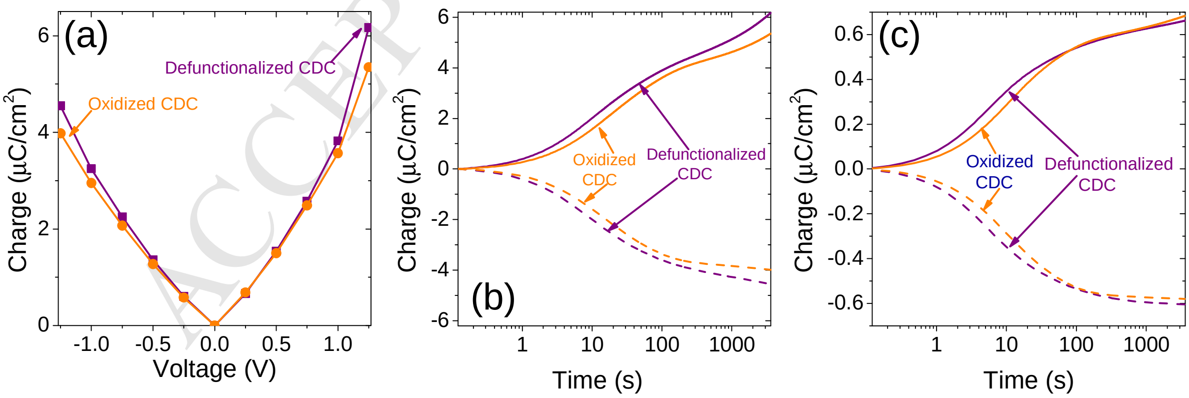

Fig. 10 – (a) Square wave amperommetry voltage-dependent charge accumulation for defunctionalized and oxidized CDCs in the positive and negative potential ranges. Time vs. charge accumulation is shown for (b) 0 → +1.25 V and 0 → -1.25 V and (c) 0 → +0.25 V and 0 → -0.25 V square wave step experiments.